Gresham's Law and the Fracture of Esoteric Alchemy

When international silver values fluctuated against the enforced Newtonian mint ratio of 1717, they triggered the mechanical arbitrage of Gresham's Law: "bad money drives out good" [@thornton1802nature]. Because gold was systematically overvalued within the English Channel—Newton's guinea priced at 21 shillings domestically while equivalent gold commanded only 20s 9d in Spain or Portugal—bullion merchants melted full-weight silver coins and shipped the raw bullion to France, India, and China, where its elemental weight commanded significantly higher purchasing power [@quinn1996competing]. An engineered mathematical disparity hollowed out the domestic money supply.

Velde's Mathematical Formalism of Bimetallic Instability

To formalize this entropic decay, modern economic historians such as François Velde define the bimetallic equilibrium via the relationship between the statutory Mint Ratio (M) and the global commodity Market Ratio (R), bounded by transaction and melting costs (c) [@velde2005bimetallism]. The condition for a functional, meta-stable bimetallic circulation is expressed mathematically as: